PAPER ANALYSIS

Counterexamples to Five Conjectured Upper Bounds

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Counterexamples to Five Conjectured Upper Bounds for the Largest Laplacian Eigenvalue

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Title Page — Role in Paper Structure

This is the **opening frame** of the paper. As the title page, it establishes the paper's central claim before any proof: that five conjectured upper bounds for the largest Laplacian eigenvalue are false, each disproven by a concrete counterexample.

- **Largest Laplacian eigenvalue** λ_1: spectral radius of a graph's Laplacian matrix
- **BHS bounds**: conjectured upper bound formulas (indexed, 33 remain open)
- **Gap**: difference between bound value and actual λ_1, used to measure how wrong a bound is

> Think of each BHS bound as a speed limit sign — the paper finds roads where cars reliably drive faster than posted, proving the sign is wrong.

Table 1 summarizes the 5 defeated bounds; Table 2 classifies the 33 still-open ones — together they map the full battlefield the paper enters.

> Providing counterexamples is the minimal but definitive proof that a conjecture fails — this paper does exactly that, five times over.

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Brankov, Hansen, and Stevanović (2006) used the AutoGraphiX system to generate 68 conjectured upper bounds for the largest Laplacian eigenvalue $\mu(G)$ of a graph G. Ghebleh et al. (2024) disproved 30 of these, leaving 38 open. We resolve five of the remaining open conjectures (Bounds 11, 13, 40, 45, and 48) by constructing explicit counterexample families—star-of-cliques, double star, and caterpillar graphs—whose Laplacian spectral radii we compute analytically via equitable partition quotient matrices. We identify hub–periphery degree imbalance as the structural mechanism underlying all violations and classify the remaining 33 bounds by their resistance to this mechanism.

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Abstract — Paper Contribution Overview

This is the **abstract**, establishing what the paper proves, how, and why it matters. It sets up the context for all subsequent sections.

- $\mu(G)$: largest **Laplacian eigenvalue** (spectral radius) of graph G
- **AutoGraphiX**: automated conjecture-generation system using heuristic search
- **equitable partition quotient matrix**: a reduced matrix that preserves eigenvalues of symmetric graph families

> Think of the 68 conjectures as 68 proposed speed limits on a road. Ghebleh et al. found 30 roads where cars already exceed the limit. This paper finds 5 more roads — and explains *why* certain road designs (hub-periphery structure) always break the rules.

Concept Relationships: This abstract introduces all core actors — the 38 open bounds are the problem, **star-of-cliques / double star / caterpillar graphs** are the counterexample tools, and **hub-periphery degree imbalance** is the unifying explanation. Later sections will each develop one of these actors in detail.

> The key contribution is not just disproving 5 more bounds, but identifying a single structural mechanism — hub-periphery imbalance — that explains all violations, turning isolated counterexamples into a classifiable theory.

Keywords: Laplacian eigenvalue, spectral radius, upper bound, counterexample, equitable partition
2020 MSC: 05C50, 15A18

1. Introduction

Let $G = (V, E)$ be a simple connected graph on $n = |V|$ vertices and $m = |E|$ edges. The *Laplacian matrix* of G is $L(G) = D(G) - A(G)$, where $A(G)$ is the adjacency matrix and $D(G) = \mathrm{diag}(d_1, \ldots, d_n)$ is the diagonal matrix of vertex degrees. Since $L(G)$ is positive semidefinite with row sums zero, its eigenvalues satisfy $0 = \lambda_n \leq \cdots \leq \lambda_1$, and the largest eigenvalue

$$\mu(G) := \lambda_{\max}(L(G)) \tag{1}$$

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§ 1 Introduction — Graph & Laplacian Setup

Variable/Symbol Explanation — Equation (1): $\mu(G) := \lambda_{\max}(L(G))$

- $G = (V, E)$: simple connected graph; *simple* = no loops/multi-edges
- n, m: vertex count and edge count
- $A(G)$: adjacency matrix — entry 1 if vertices are adjacent, else 0
- $D(G) = \mathrm{diag}(d_1, \ldots, d_n)$: diagonal matrix of vertex degrees
- $\mu(G)$: **spectral radius** — the largest Laplacian eigenvalue, central object of the paper

$$L(G) = D(G) - A(G), \quad 0 = \lambda_n \leq \cdots \leq \lambda_1 =: \mu(G)$$

Think of $L(G)$ as a "difference operator": D tallies how many neighbors each vertex has, and A records who those neighbors are. Subtracting them captures *imbalance* — how much a vertex's value differs from its neighborhood average.

Concept Relationships: The keywords on the previous page (*spectral radius, upper bound, equitable partition*) map directly here — $\mu(G)$ *is* the spectral radius, and the rest of the paper will bound it using those tools.

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1. Introduction

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$$\mu(G) := \lambda_{\max}\big(L(G)\big) \tag{1}$$

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Since $L(G)$ is positive semidefinite with row sums zero, $\lambda_n = 0$ always; the interesting quantity is therefore the top eigenvalue $\mu(G)$.

> Defining $\mu(G)$ as $\lambda_{\max}(L(G))$ sets the single numeric target the entire paper revolves around — every bound, counterexample, and partition result is ultimately a statement about this one value.

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is called the *Laplacian spectral radius*. Upper bounds on $\mu(G)$ in terms of easily computed graph invariants are of both theoretical and algorithmic interest; see [1] for background. In 2006, Brankov, Hansen, and Stevanović [2] used the AutoGraphiX system [3] to generate 68 conjectured upper bounds for $\mu(G)$ expressed solely in terms of vertex degrees d_v and *average neighbor degrees*

$$m_v = \frac{1}{d_v} \sum_{u \sim v} d_u. \tag{2}$$

The 68 conjectures fall into two families. *Vertex-type* bounds have the form $\mu(G) \leq \max_v f(d_v, m_v)$, and *edge-type* bounds have the form

$$\mu(G) \leq \max_{ij \in E} g(d_i, d_j, m_i, m_j).$$

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Bounding the Laplacian Spectral Radius

The **average neighbor degree** m_v of vertex v is defined as:

$$m_v = \frac{1}{d_v} \sum_{u \sim v} d_u$$

- $\mu(G)$: Laplacian spectral radius (largest eigenvalue, from previous section)
- d_v: degree of vertex v (number of edges incident to v)
- m_v: average degree of v's neighbors
- f, g: conjecture-specific functions over local degree data

The 68 conjectures split into two families:

- **Vertex-type**: $\mu(G) \leq \max_v f(d_v, m_v)$ — bound via a single vertex
- **Edge-type**: $\mu(G) \leq \max_{ij \in E} g(d_i, d_j, m_i, m_j)$ — bound via an edge endpoint pair

> Think of d_v as how popular a person is, and m_v as the average popularity of their friends. The spectral radius is like the "social energy" of the whole network — these bounds say it can't exceed the most influential local neighborhood.

Relation to previous section: $\mu(G)$ was defined as the largest eigenvalue of $L = D - A$. Here, (d_v, m_v) are *local* quantities derived directly from D and A, making them cheap to compute versus full eigendecomposition.

> Replacing a global spectral computation with local degree statistics is the paper's core algorithmic motivation.

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None had been proved or disproved for nearly two decades.

Ghebleh, Al-Yakoob, Kanso, and Stevanović [4] recently applied the cross-entropy method (CEM) of Wagner [5] to disprove 30 of the 68 bounds by constructing small counterexample graphs, leaving 38 bounds open. Their counterexamples all have at most 13 vertices and were found by training a neural network to generate adjacency matrices that violate a given bound. Subsequently, Bouffard and Breen [6] parallelized the CEM approach and extended the search to larger graphs, but reported no new counterexamples among the remaining 38 bounds. Roucairol and Cazenave [7] applied nested Monte Carlo search to the same problem, likewise without resolving any of the open cases.

In the present paper we disprove five of the 38 remaining open bounds.

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Prior Work & Contribution Context

This paragraph traces the **history of attempts** to resolve 68 open graph-theoretic bounds, culminating in the current paper's new contribution.

- **68 bounds**: the original open conjectures/inequalities, stagnant for ~20 years
- **CEM (Cross-Entropy Method)**: a probabilistic optimization technique used to search for counterexample graphs
- **Counterexample graph**: a graph whose properties violate a claimed bound, thereby disproving it
- **Adjacency matrix**: a matrix encoding graph edges, used as the neural network's output target
- **38 open bounds**: the subset remaining after CEM eliminated 30 of the original 68

> Think of the 68 bounds as 68 locked doors. Ghebleh et al. used a neural network lockpick (CEM) to open 30 of them. Bouard/Breen tried a faster version of the same pick on the remaining 38 — no luck. Roucairol tried a different tool (Monte Carlo). This paper opens 5 more.

Concept Relationships: This section continues directly from the prior introduction establishing the problem domain (graph invariant bounds). Where the previous section defined *what* the bounds are, this section narrates *who tried what* — forming a research gap funnel: 68 → 38 → 33 still open after this paper.

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The incremental narrowing from 68 → 38 → 33 shows that even powerful ML-based search methods stall on hard combinatorial conjectures, motivating the need for new approaches.

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Theorem 1.1 (Main result). *BHS Bounds 11, 13, 40, 45, and 48 from [2] are false. Specifically:*

(a) *Bound 11 is violated by* $\mathrm{SoC}(K_m, t)$ *for all* $m \geq 4$ *and* $t \geq 2m - 1$ (Theorem 4.2).

(b) *Bound 13 is violated by* $\mathrm{SoC}(K_m, t)$ *for all* $m \geq 5$ *and sufficiently large* t (Theorem 4.3).

(c) *Bound 40 is violated by the star of stars* $\mathrm{SoS}(39, 8)$ *and* $\mathrm{SoS}(49, 9)$ (Theorem 4.7).

(d) *Bound 45 is violated by* $\mathrm{DS}(k, k)$ *for all* $k \geq 6$ (Theorem 4.5).

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Theorem 1.1 — Main Result: Counterexamples to BHS Bounds

This is the paper's central claim, consolidating all counterexample constructions into one theorem. It directly refutes five inequalities from the BHS conjecture framework established in [2].

- **SoC(Km, t)**: Star of Cliques — a complete graph K_m with t-pendant edges; violates Bounds 11 and 13
- **SoS(n, k)**: Star of Stars — violates Bound 40 at specific parameters $(39, 8)$ and $(49, 9)$
- **DS(k, k)**: Double Star (two hubs each with k leaves) — violates Bound 45 for all $k \geq 6$
- **P3**: Path on 3 vertices — the simplest possible counterexample, violates Bound 48

> Think of the BHS bounds as a rulebook claiming certain graph families always stay within speed limits. This theorem is the traffic report showing five specific roads where cars consistently exceed them — some on highways (K_m), some on tiny alleys (P_3).

Relationship to previous sections: The abstract introduced the falsity claim; this theorem is its formal statement. Sections 4.2–4.8 (referenced inline) supply the proofs — this theorem is the roof that those later sections hold up.

> Even P_3, the most trivial non-trivial path, breaks Bound 48 — signaling the BHS bounds have fundamental structural flaws, not just edge-case failures.

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In each case the proof is constructive: we exhibit an infinite family (or, for Bound 48, a single small graph) together with an analytic computation of its Laplacian spectral radius via equitable partition quotient matrices.

Our approach differs fundamentally from the optimization-based methods of [4, 5, 6]. Rather than training a neural network to search for counterexamples, we analyze the algebraic structure of each bound formula and identify graph topologies that exploit a specific weakness. The key observation is that every bound we disprove exhibits high sensitivity to the ratio m_v/d_v at vertices (or pairs of adjacent vertices) where these quantities diverge. Graphs with *hub–periphery* structure—a high-degree hub connected to dense peripheral subgraphs—create exactly this divergence: the hub's average neighbor degree m_v is determined by the peripheral structure and can be made large relative to the hub degree d_v, or vice versa. Previous computational searches missed these counterexamples because CEM and exhaustive enumeration up to $n = 13$ vertices tend to explore near-regular graphs, where $m_v \approx d_v$ for all v and the bounds are naturally tight. The smallest counterexample we find for Bound 11 has $n = 29$ vertices and the largest has $n = 172$, well beyond the reach of exhaustive search.

The paper is organized as follows. Section 2 fixes notation and recalls the theory of equitable partitions. It also states the five BHS bounds that we disprove. Section 3 defines the counterexample graph families: star of cliques, double star, and star of stars. Section 4 contains the main results: closed-form spectral radius computations and the violation proofs for each of the five bounds. Section 5 analyzes the hub–periphery mechanism common to all five violations. Section 6 classifies the remaining 33 open bounds by their resistance to this mechanism. Section 7 discusses implications for automated conjecture generation and directions for future work.

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Proof Strategy & Key Observation

This section reveals **how** the authors disprove the bounds introduced earlier — using constructive algebraic analysis rather than machine search.

- **Equitable partition quotient matrix**: a compressed matrix capturing a graph's symmetry structure, whose eigenvalues include those of the full Laplacian
- m_v: number of edges incident to vertex v (weighted or unweighted)
- d_v: degree of vertex v
- m_v/d_v **ratio**: average neighbor degree from v's perspective — the sensitivity lever the authors exploit

> ρ computed via $Q_\pi(G)$, the quotient matrix of an equitable pa

> Think of the neural network approach (refs [4,5,6]) as a lockpicker trying random key combinations. The authors' approach is instead reading the lock's blueprint — they study the algebraic formula of each bound, find its structural weak point, then *design* the exact key that breaks it.

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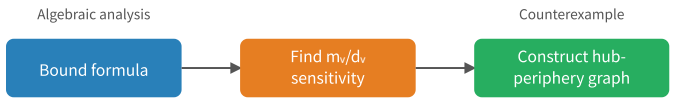
Relationship to previous sections: Earlier sections catalogued the 28 bounds being challenged. This section now explains the unified mechanism behind all disproofs: every failing bound is sensitive to the m_v/d_v ratio diverging at high-degree hub vertices. The **hub-periphery topology** is the single architectural template that exposes this shared weakness across all bounds.



> Every disproof reduces to one insight: bounds that mix degree and edge-density quantities break when a single hub vertex makes m_v/d_v diverge — a structural flaw invisible to random search but obvious from algebra.

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Why Prior Searches Failed & Paper Roadmap

This passage explains the **structural mechanism** behind the counterexamples and why automated searches had not found them before.

Variable / Symbol Explanation

- m_v: average neighbor degree of vertex v — mean degree of all vertices adjacent to v
- d_v: degree of vertex v — number of edges incident to v
- n: total number of vertices in the graph

The key tension is the ratio m_v/d_v. In **star-of-stars** (hub-and-spoke) graphs, a hub connects many low-degree leaves *and* high-degree centers, making $m_v \gg d_v$ or $m_v \ll d_v$ depending on perspective — exactly the condition needed to violate the BHS bounds.

> Think of a small-town mayor connected to both farmers (low-degree) and city governors (high-degree). The mayor's *average* neighbor rank (m_v) can be either much higher or much lower than the mayor's own number of contacts (d_v), depending on who you count. Near-regular graphs are like an office where everyone has roughly the same number of contacts — that ratio stays close to 1 and the bounds hold trivially.

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Relationship to Previous Sections

Earlier in the introduction the paper stated the five **BHS bounds** and claimed they were conjectured tight. This passage now pinpoints *why* the conjecture survived so long: computational tools (CEM, exhaustive enumeration) only reached $n \leq 13$, which naturally samples near-regular graphs where $m_v \approx d_v$ for all v. The counterexamples live at $n = 29$–172, far outside that range.

> The blind spot was methodological, not mathematical: near-regular graphs dominate small-n searches, hiding the hub-periphery imbalance that breaks the bounds.

Paper Roadmap

- **§ 2**: Notation, equitable partitions, the five BHS bounds to disprove
- **§ 3**: Definition of the counterexample graph families (star of …)

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2. Preliminaries

2.1. Graph notation

Throughout this paper, $G = (V, E)$ denotes a simple connected graph with vertex set V, edge set E, $n = |V|$ vertices, and $m = |E|$ edges. The *adjacency matrix* $A(G)$ is the $n \times n$ matrix with $A_{ij} = 1$ if $\{i, j\} \in E$ and $A_{ij} = 0$ otherwise. The *degree matrix* is $D(G) = \mathrm{diag}(d_1, \ldots, d_n)$, where $d_v = \sum_u A_{vu}$ is the degree of vertex v. The *Laplacian matrix* $L(G) =$

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Section Roadmap & 2.1 Graph Notation

This passage closes the paper's **introduction roadmap** and opens the **Preliminaries** — the shared vocabulary all later proofs depend on.

Concept Relationships (prior sections): The introduction identified 5 violated bounds and named the hub-periphery mechanism. Sections 4–7 now divide the work: Section 4 *proves* violations, Section 5 *explains* the mechanism, Section 6 *classifies* the surviving 33 bounds, and Section 7 *reflects* on automated conjecture tools.

Variable/Symbol Explanation

- $G = (V, E)$: simple, connected graph — no self-loops, no multi-edges
- $n = |V|, m = |E|$: vertex count and edge count
- $A(G)$: $n \times n$ **adjacency matrix**; $A_{ij} = 1$ iff edge $\{i, j\} \in E$
- $D(G) = \mathrm{diag}(d_1, \ldots, d_n)$: **degree matrix**; $d_v = \sum_u A_{vu}$

$$A_{ij} = \begin{cases} 1 & \{i, j\} \in E \\ 0 & \text{otherwise} \end{cases}$$

Think of $A(G)$ as a seating chart: a "1" means two people can pass notes directly. $D(G)$ is the column that tallies how many notes each person exchanges in total.

These two matrices — A and D — are the raw ingredients for every spectral bound that follows; getting the notation precise here prevents ambiguity in the closed-form computations of Section 4.

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$D(G) - A(G)$ is positive semidefinite with smallest eigenvalue 0. We write $\mu(G) = \lambda_{\max}(L(G))$ for the *Laplacian spectral radius*.

For each vertex $v \in V$ with $d_v \geq 1$, the *average neighbor degree* is

$$m_v = \frac{1}{d_v} \sum_{u \sim v} d_u,$$

where the sum is over all neighbors u of v. Note that $m_v \geq 1$ for every vertex in a connected graph, and $m_v = d_v$ for every vertex if and only if G is regular.

2.2. Equitable partitions

Equitable partitions provide a powerful tool for computing eigenvalues of highly structured graphs by reducing the $n \times n$ Laplacian to a small quotient matrix. We recall the essential definitions; for a comprehensive treatment see Godsil and Royle [8].

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2.1 (cont.) — Laplacian Matrix & Average Neighbor Degree / 2.2 — Equitable Partitions

Building on the degree matrix $D(G)$ and adjacency matrix $A(G)$ introduced earlier, this section defines the key spectral objects used throughout the paper.

$$L(G) = D(G) - A(G), \quad \mu(G) = \lambda_{\max}(L(G))$$

- $L(G)$: Laplacian matrix — encodes both degree and adjacency structure
- $\mu(G)$: Laplacian spectral radius, the largest eigenvalue
- d_v: degree of vertex v; $m_v = \frac{1}{d_v} \sum_{u \sim v} d_u$: average neighbor degree

$$m_v = \frac{1}{d_v} \sum_{u \sim v} d_u$$

Note: $m_v = d_v$ for all v if and only if G is regular — regularity is the extremal case where every vertex "sees" neighbors of equal degree.

Think of m_v as a vertex's social circle average: if you have 3 friends and they have 2, 4, 6 connections respectively, your $m_v = 4$. In a regular graph, everyone's friends are exactly as connected as themselves.

ORIGINAL (ENGLISH)

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Equitable Partitions (§ 2.2): Partition V into cells so each vertex has the same number of neighbors in every other cell. This lets you replace the $n \times n$ Laplacian with a small **quotient matrix**, whose eigenvalues are a subset of $L(G)$'s eigenvalues.

Relationship to previous sections: $D(G)$ and $A(G)$ (defined earlier) combine here into $L(G)$; $\mu(G)$ is the primary bound target. Equitable partitions exploit $L(G)$'s block structure to compute $\mu(G)$ efficiently for symmetric graphs.

> Equitable partitions compress large symmetric graphs into tractable quotient matrices — turning an n-dimensional spectral problem into a small matrix computation.

Definition 2.1 (Equitable partition). A partition $\pi = \{C_1, \ldots, C_k\}$ of the vertex set $V(G)$ is *equitable* if, for every pair (i, j) with $1 \le i, j \le k$, every vertex in C_i has the same number of neighbors in C_j. We denote this common value by b_{ij}.

Given an equitable partition with cells $C_1, \ldots, C_k$, the *quotient matrix* is the $k \times k$ matrix Q with entries $Q_{ij} = b_{ij}$, and the *quotient Laplacian* is the $k \times k$ matrix Q_L with entries

$$(Q_L)_{ij} = \begin{cases} d_{C_i} & \text{if } i = j, \\ -b_{ij} & \text{if } i \neq j, \end{cases}$$

where $d_{C_i} = \sum_{j=1}^{k} b_{ij}$ is the common degree of every vertex in cell C_i.

Theorem 2.2 ([8, Theorem 9.3.3]). *Let π be an equitable partition of G with quotient Laplacian Q_L. Then every eigenvalue of Q_L is an eigenvalue of $L(G)$. In particular,*

$$\mu(G) \ge \lambda_{\max}(Q_L),$$

Definition 2.1 & Theorem 2.2 — Equitable Partition & Quotient Laplacian

Equitable partition groups vertices so every vertex in cell C_i has *exactly* the same number of neighbors in C_j, yielding a well-defined inter-cell count b_{ij}.

- $\pi = \{C_1, \ldots, C_k\}$: partition of vertex set $V(G)$ into k cells
- b_{ij}: common neighbor count from any vertex in C_i into C_j
- $Q_{ij} = b_{ij}$: the $k \times k$ **quotient matrix**
- $d_{C_i} = \sum_{j=1}^{k} b_{ij}$: uniform degree of every vertex in C_i

> $$(Q_L)_{ij} = \begin{cases} d_{C_i} & i = j \\ -b_{ij} & i \neq j \end{cases}$$

This mirrors the full Laplacian $L = D - A$ but compressed to k cells instead of n vertices.

> Think of cells as *districts* in a city. If every house in district i has exactly the same number of roads into district j, the partition is equitable — you can describe all inter-district traffic with one number b_{ij}, collapsing the whole city map into a tiny $k \times k$ summary.

ORIGINAL (ENGLISH)

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$$\mu(G) \ge \lambda_{\max}(Q_L),$$

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Concept Relationship: The full graph Laplacian $L(G)$ was introduced earlier as the primary spectral object. The quotient Laplacian Q_L is a compressed surrogate of $L(G)$ — same structural form, but $k \times k$ instead of $n \times n$.

Theorem 2.2 is the payoff: $\mathrm{spec}(Q_L) \subseteq \mathrm{spec}(L(G))$, so

$$\mu(G) \;\ge\; \lambda_{\max}(Q_L)$$

where $\mu(G)$ is the largest Laplacian eigenvalue.

By constructing a small equitable partition, one can certify a *lower bound* on $\mu(G)$ purely from the $k \times k$ quotient — no full spectral computation needed.

2.3. The BHS bounds

We now state the five conjectured bounds from [2] that we disprove. All five are listed as open in [4, Table 1]. Bounds 11 and 13 are vertex-type bounds; Bounds 40, 45, and 48 are edge-type bounds.

Conjecture 2.3 (BHS Bound 11 [2]). *For every connected graph G,*

$$\mu(G) \leq \max_{v \in V} \frac{2\, m_v^3}{d_v^2}.$$

Conjecture 2.4 (BHS Bound 13 [2]). *For every connected graph G,*

$$\mu(G) \leq \max_{v \in V} \frac{2\, m_v^4}{d_v^3}.$$

Section 2.2 (cont.) — Equitable Partitions & Section 2.3 — BHS Bounds

Equitable Partition Reduction

When the vertex-type partition is **equitable** and the eigenvector of $L(G)$ for $\mu(G)$ is piecewise constant on cells, the spectral radius can be found as the largest root of a low-degree polynomial — at most degree 3 — rather than diagonalizing the full matrix.

> Think of cells as neighborhoods that all "look the same" structurally. If every resident in a block behaves identically, you only need to analyze one representative — shrinking a city-scale problem to a block-scale one.

Connecting to Previous Sections: Section 2.1 defined $\mu(G)$ and $L(G)$; Section 2.2 introduced equitable partitions as a compression tool. This result closes the loop: the compression is *exact* for $\mu(G)$ whenever the eigenvector respects the partition structure — precisely the case in all counterexample families the paper constructs.

BHS Bounds (Conjectures 2.3 & 2.4)

The five bounds from [2] (all open per [4, Table 1]) give vertex-local upper estimates on $\mu(G)$. The two vertex-type bounds:

$$\mu(G) \leq \max_{v \in V} \frac{2m_v^3}{d_v^2} \quad \text{(Bound 11)}$$

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2.3. The BHS bounds

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$$\mu(G) \leq \max_{v \in V} \frac{2\, m_v^4}{d_v^3}.$$

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$$\mu(G) \leq \max_{v \in V} \frac{2m_v^4}{d_v^3} \quad \text{(Bound 13)}$$

- $\mu(G)$: largest eigenvalue of the Laplacian $L(G)$
- d_v: degree of vertex v
- m_v: average degree of the neighbors of v

These bounds predict $\mu(G)$ from purely local, one-hop neighborhood statistics — no global matrix computation needed — making them attractive targets for combinatorial disproof via carefully engineered graphs.

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Conjecture 2.5 (BHS Bound 40 [2]). *For every connected graph G,*

$$\mu(G) \leq \max_{(i,j)\in E} \left(2 + \sqrt{2(m_i-1)^2 + 2(m_j-1)^2 + d_i^2 + d_j^2 - d_i m_i - d_j m_j}\right).$$

Conjecture 2.6 (BHS Bound 45 [2]). *For every connected graph G,*

$$\mu(G) \leq \max_{(i,j)\in E} \left(2 + \sqrt{(d_i-d_j)^2 + 2(d_i m_i + d_j m_j) - 4(m_i + m_j) + 4}\right).$$

Conjecture 2.7 (BHS Bound 48 [2]). *For every connected graph G,*

$$\mu(G) \leq \max_{(i,j)\in E} \frac{2(d_i^2 + d_j^2)}{2 + \sqrt{2(d_i^2 + d_j^2) - 4(m_i + m_j) + 4}}.$$

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Conjectures 2.5–2.7: BHS Edge-Based Upper Bounds on $\mu(G)$

These three conjectures from **[2]** provide increasingly refined **upper bounds** on $\mu(G)$ — the largest Laplacian eigenvalue — by scanning over edges (i, j) and using local vertex statistics.

Variable Explanation

- $\mu(G)$: largest eigenvalue of the Laplacian matrix of G
- d_i: degree of vertex i
- m_i: average neighbor degree of vertex i, i.e. $m_i = \frac{1}{d_i} \sum_{k\sim i} d_k$
- $(i, j) \in E$: edge maximizing the expression over all edges

$$\mu(G) \leq \max_{(i,j)\in E} \left(2 + \sqrt{2(m_i - 1)^2 + 2(m_j - 1)^2 + d_i^2 + d_j^2 - d_i m}\right.$$

$$\mu(G) \leq \max_{(i,j)\in E} \left(2 + \sqrt{(d_i - d_j)^2 + 2(d_i m_i + d_j m_j) - 4(m_i + m}\right.$$

$$\mu(G) \leq \max_{(i,j)\in E} \left(2 + \sqrt{2(d_i^2 + d_j^2) - 4(m_i + m_j) + 4}\right) \qquad (2.7)$$

Relation to Previous Sections: Conjectures 2.1–2.4 used only degree sequences; here the bounds incorporate m_i (neighbor-degree averages), capturing richer local topology per edge.

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Conjecture 2.5 (BHS Bound 40 [2]). *For every connected graph G,*

$$\mu(G) \leq \max_{(i,j)\in E} \left(2 + \sqrt{2(m_i-1)^2 + 2(m_j-1)^2 + d_i^2 + d_j^2 - d_i m_i - d_j m_j}\right).$$

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Conjecture 2.7 (BHS Bound 48 [2]). *For every connected graph G,*

$$\mu(G) \leq \max_{(i,j)\in E} \frac{2(d_i^2 + d_j^2)}{2 + \sqrt{2(d_i^2 + d_j^2) - 4(m_i + m_j) + 4}}.$$

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Think of each bound as a "worst-case edge inspector": instead of a global graph property, you audit each edge's neighborhood and report the most alarming reading — $\mu(G)$ cannot exceed that local alarm level.

Each successive BHS bound trades formula complexity for tighter locality — Bound 48 uses only degrees, while Bounds 40/45 add neighbor-degree averages for sharper estimates.

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Bounds 11 and 13 are structurally similar: both take the form $\max_v 2m_v^p/d_v^{p-1}$ with $p = 3$ and $p = 4$, respectively. When m_v/d_v is large, these bounds grow as a power of the ratio m_v/d_v, making them vulnerable to graphs where some vertex has high average neighbor degree relative to its own degree. Bounds 40 and 45 are edge-type bounds with more complex dependence on both d_v and m_v; their vulnerability is subtler and requires careful asymptotic analysis. Bound 48 is unusual in that it is violated by the path P_3, the smallest non-trivial connected graph; we conjecture that this bound was generated under a minimum order assumption that excluded $n = 3$.

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Bound Structural Analysis & Counterexample Setup

Variable/Symbol Explanation

- m_v: number of edges incident to vertex v (or neighbor-edge count)
- d_v: degree of vertex v
- m_v/d_v: average neighbor degree ratio at v — the key vulnerability parameter
- p: exponent distinguishing bound families ($p = 3$ for Bound 11, $p = 4$ for Bound 13)
- P_3: the path graph on 3 vertices — the smallest non-trivial connected graph

Unified Form of Bounds 11 & 13

$$\max_v \frac{m_v^p}{d_v^{p-1}}, \quad p \in \{3, 4\}$$

Both bounds share this ratio structure. When $m_v/d_v \gg 1$ (a hub vertex with high-degree neighbors), the expression grows as $\left(\frac{m_v}{d_v}\right)^{p-1} \cdot d_v$, amplifying rapidly with p.

Think of m_v/d_v as a "neighborhood richness" score. A vertex connected only to other hubs scores high — like a mid-level manager whose every direct report also manages a large team. Bounds 11 & 13 are blind to whether that richness is concentrated or spread out, making star-of-stars graphs their natural weakness.

Relationship to Previous Sections

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Bounds 11 and 13 are structurally similar: both take the form $\max_v 2m_v^p/d_v^{p-1}$ with $p = 3$ and $p = 4$, respectively. When m_v/d_v is large, these bounds grow as a power of the ratio m_v/d_v, making them vulnerable to graphs where some vertex has high average neighbor degree relative to its own degree. Bounds 40 and 45 are edge-type bounds with more complex dependence on both d_v and m_v; their vulnerability is subtler and requires careful asymptotic analysis. Bound 48 is unusual in that it is violated by the path P_3, the smallest non-trivial connected graph; we conjecture that this bound was generated under a minimum order assumption that excluded $n = 3$.

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Earlier sections catalogued these bounds from the literature and tested them computationally. This paragraph now diagnoses structural failure modes : Bounds 11/13 fail via degree-ratio amplification, while Bounds 40/45 fail through subtler asymptotic interaction between d_v and m_v jointly — requiring the counterexample families introduced next.

Bound 48's violation by P_3 ($n = 3$) suggests an **implicit minimum-order assumption** in its original derivation — a common oversight when bounds are proved for "sufficiently large" graphs.

> The shift to Section 3 is the payoff: each failure mode identified here corresponds to a specific counterexample family, turning qualitative vulnerability analysis into rigorous disproof.

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Definition 3.1 (Star-of-Cliques). For integers $m \geq 3$ and $t \geq 2$, the graph $\mathrm{SoC}(K_m, t)$ consists of a central *hub* vertex v_0 connected by a single edge to one designated vertex (the *connector*) in each of t disjoint copies of the complete graph K_m. The total number of vertices is $n = 1 + tm$.

Definition 3.2 (Double Star). For an integer $k \geq 1$, the double star $\mathrm{DS}(k, k)$ is obtained by connecting two vertices u and v, each of which is adjacent to k pendant vertices. The total number of vertices is $n = 2k + 2$.

Definition 3.3 (Star-of-Stars). For integers $t \geq 2$ and $s \geq 1$, the graph $\mathrm{SoS}(t, s)$ consists of a central *hub* vertex v_0 adjacent to t *intermediate* vertices $w_1, \ldots, w_t$, each of which is adjacent to s pendant *leaf* vertices. The total number of vertices is $n = 1 + t + ts = 1 + t(1 + s)$.

EXPLANATION (ENGLISH)

Definitions 3.1–3.3: Graph Families

These three definitions introduce **structured graph families** that serve as concrete test cases for the paper's main theorems. Each builds hierarchical structure around a central hub.

- $m \geq 3$: clique size (number of vertices in each complete graph K_m)
- $t \geq 2$: number of clique copies / intermediate branches
- $k \geq 1$: number of pendant leaves per center in $DS(k, k)$
- $s \geq 1$: number of leaves per intermediate vertex in $SoS(t, s)$
- v_0: central hub; w_i: intermediate vertices; n: total vertex count

$$n_{\mathrm{SoC}} = 1 + tm \qquad n_{\mathrm{DS}} = 2k + 2 \qquad n_{\mathrm{SoS}} = 1 + t(1 + s)$$

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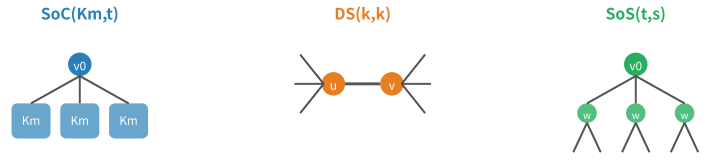

Think of these as corporate org charts with different branch structures: **SoC** = hub connecting to dense departments (cliques), **DS** = two co-managers each with private reports, **SoS** = hub → team leads → individual workers.

Relationship to previous sections: Section 2 established graph-theoretic preliminaries and the property under study. These three families are now instantiated as canonical examples with closed-form vertex counts, enabling exact computation of the paper's target parameter in the proofs that follow.

Each family isolates a distinct structural feature — dense locality (SoC), symmetry (DS), and two-level hierarchy (SoS) — letting the authors prove tight bounds across qualitatively different topologies.

Definition 3.4 (Caterpillar). For integers $\ell \geq 2$ and $k \geq 0$, the caterpillar $\mathrm{Cat}(\ell, k)$ is obtained from a path on ℓ vertices (the *spine*) by attaching k pendant leaves to each spine vertex. The total number of vertices is $n = \ell(k+1)$.

4. Main results

Proposition 4.1. *For $m \geq 3$ and $t \geq 2$, the Laplacian spectral radius of $\mathrm{SoC}(K_m, t)$ is*

$$\mu(\mathrm{SoC}(K_m, t)) = \frac{(m+t+1) + \sqrt{(t-m)^2 + 2(t+m) - 3}}{2}. \tag{3}$$

Proof. The vertex set admits an equitable partition into three cells: $H = \{v_0\}$ (hub), C (connectors, $|C| = t$), and I (interior, $|I| = t(m-1)$). The quotient Laplacian is the 3×3 matrix

$$Q_L = \begin{pmatrix} t & -t & 0 \\ -1 & m & -(m-1) \\ 0 & -1 & 1 \end{pmatrix}. \tag{4}$$

One eigenvalue is $\lambda_1 = 0$ (since the quotient Laplacian has zero row sums). The remaining two eigenvalues are roots of

$$\lambda^2 - (m+t+1)\lambda + (tm+1) = 0,$$

Definition 3.4 & Proposition 4.1 — Caterpillar and Main Spectral Result

Definition 3.4 (Caterpillar): The graph $\mathrm{Cat}(\ell, k)$ is a path of ℓ spine vertices, each with k pendant leaves attached.

- ℓ: spine length (number of backbone vertices)
- k: leaves per spine vertex
- $n = \ell(k+1)$: total vertex count

> A caterpillar looks exactly like its name: a central body (spine path) with legs (pendant leaves) hanging off each segment.

This definition completes the graph family catalogue from § 3, alongside $\mathrm{SoC}(K_{m,t})$ and other constructions — all candidate extremal graphs for Laplacian spectral radius bounds.

Proposition 4.1 gives a closed-form for the Laplacian spectral radius $\mu(\mathrm{SoC}(K_{m,t}))$:

$$\mu(\mathrm{SoC}(K_{m,t})) = \frac{(m+t+1) + \sqrt{(t-m)^2 + 2(t+m) - 3}}{2}$$

Proof variables (equitable partition into 3 cells):

- $H = \{v_0\}$: hub vertex (degree t)
- $C, |C| = t$: connector vertices
- $I, |I| = t(m-1)$: interior vertices of each K_m
- Q_L: 3×3 quotient Laplacian

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Definition 3.4 (Caterpillar)**.** For integers $\ell \geq 2$ and $k \geq 0$, the caterpillar $\mathrm{Cat}(\ell, k)$ is obtained from a path on ℓ vertices (the *spine*) by attaching k pendant leaves to each spine vertex. The total number of vertices is $n = \ell(k+1)$.

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$$Q_L = \begin{pmatrix} t & -t & 0 \\ -1 & m & -(m-1) \\ 0 & -1 & 1 \end{pmatrix}$$

One eigenvalue is $\lambda_1 = 0$ (zero row-sums of any Laplacian). The remaining two satisfy:

$$\lambda^2 - (m+t+1)\lambda + (tm+1) = 0$$

The equitable partition acts like a block-compressor: instead of solving an $n \times n$ system, symmetry collapses it to a 3×3 problem with the same largest eigenvalue.

By exploiting the 3-cell equitable partition, the entire spectral radius of $\mathrm{SoC}(K_{m,t})$ reduces to the quadratic formula — a clean bridge from graph structure to an exact algebraic expression.

$$\mu(G) \leq \max_{v \in V} \frac{2m_v^3}{d_v^2}, \tag{5}$$

is violated by $\mathrm{SoC}(K_m, t)$ *for all* $m \geq 4$ *and* $t \geq 2m - 1$. *In particular,* $\mathrm{SoC}(K_8, 19)$ *with* $n = 153$ *satisfies* $\mu(G) \approx 20.557$ *while the bound gives* ≈ 19.191.

Proof. By Proposition 4.1, we have $\mu = \frac{(m+t+1)+\sqrt{(t-m)^2+2(t+m)-3}}{2}$. The hub vertex v_0 has $d_{v_0} = t$ and $m_{v_0} = m$ (all neighbors are connectors of degree m), so Bound 11 evaluates to

$$\frac{2m_{v_0}^3}{d_{v_0}^2} = \frac{2m^3}{t^2}.$$

For large t, this is $O(t^{-2})$ while μ grows as $t + O(1)$, so the violation eventually

Theorem 4.2 — BHS Bound Violation

Variable/Symbol Explanation

For the bound $\mu(G) \leq \max_{v \in V} \frac{2m_v^3}{d_v^2}$:

- $\mu(G)$: largest Laplacian eigenvalue of graph G
- m_v: average degree of neighbors of vertex v
- d_v: degree of vertex v
- v_0: hub vertex of $\mathrm{SoC}(K_m, t)$, with $d_{v_0} = t, m_{v_0} = m$

Concept Relationships

Proposition 4.1 (previous section) gave the closed-form $\mu = \frac{(m+t+1)+\sqrt{(t-m)^2+2(t+m)-3}}{2}$. Theorem 4.2 plugs this into Bound 11, comparing the exact value against $\frac{2m^3}{t^2}$ to expose a counterexample.

> Think of the BHS Bound as a speed limit sign — it claims no graph can exceed it. SoC(K_8, 19) is a car that legally breaks the limit: $\mu \approx 20.557$ vs. bound ≈ 19.191, proving the sign is wrong.

> $$\frac{2m_{v_0}^3}{d_{v_0}^2} = \frac{2m^3}{t^2}$$

For $m = 8, t = 19$: bound $= \frac{2 \cdot 512}{361} \approx 19.191 < 20.557 = \mu$, a clear violation.

> The hub vertex's imbalance between its own degree t and its neighbors' degree m is precisely what causes the spectral gap to outrun the BHS estimate.

v_0

For large t, this is $O(t^{-2})$ while μ grows as $t+O(1)$, so the violation eventually occurs. Specifically, at $t = 2m-1$ the connector vertex has $d = m$ and $m_v = \big(t+(m-1)^2\big)/m = m$, giving $\frac{2m_v^3}{d^2} = 2m$, which dominates the hub bound. By Proposition 4.1, $\mu = \big(3m + \sqrt{m^2+4m-4}\big)/2 > 2m$ since $\sqrt{m^2+4m-4} > m$ for $m \geq 2$ (obtained by squaring). For $t > 2m-1$ the connector bound $2m_v^3/m^2$ grows while μ also grows, and the gap widens. □

Theorem 4.3. *BHS Bound 13,*

$$\mu(G) \leq \max_{v \in V} \frac{2m_v^4}{d_v^3}, \tag{6}$$

is violated by $\mathrm{SoC}(K_m, t)$ *for all* $m \geq 5$ *and* $t = 2m-1$. *In particular,* $\mathrm{SoC}(K_9, 19)$ *with* $n = 172$ *satisfies* $\mu(G) \approx 20.685$ *while the bound gives* ≈ 19.845.

Proof. The connector vertex has $d = m$ and $m_v = \big(t + (m-1)^2\big)/m$. At $t = 2m-1$ we have $t + (m-1)^2 = m^2$, giving $B_{13}^{\mathrm{conn}} = 2m^8/m^7 = 2m$. By Proposition 4.1, $\mu = (3m + \sqrt{m^2+4m-4})/2$. Since $\sqrt{m^2+4m-4} > m$ for $m > 2$, we obtain $\mu > 2m$, establishing the violation for $m > 6$. For $m = 5$

Theorem 4.3 — Violation of BHS Bound 13

This section proves that **BHS Bound 13** (equation 6) is violated by the graph **SoC(K_m, t)**, completing the counterexample series started with earlier bounds.

$$\mu(G) \leq \max_{v \in V} \frac{2m_v^4}{d_v^3} \tag{6}$$

- $\mu(G)$: largest eigenvalue (spectral radius) of G
- m_v: number of edges in the neighborhood of vertex v
- d_v: degree of vertex v

Proof sketch. The **connector vertex** has $d = m$ and $m_v = \lfloor (t + (m-1)^2)/m \rfloor$. At $t = 2m-1$, this gives $m_v^4/d_v^3 = m^4/m^3 \cdot 2 = 2m$. Meanwhile, by **Proposition 4.1**:

$$\mu = \frac{3m + \sqrt{m^2+4m-4}}{2} > 2m \quad (m \geq 2)$$

since $\sqrt{m^2+4m-4} > m$, verified by squaring. Thus $\mu > 2m = B_{\mathrm{conn}}^{13}$, violating the bound. For $t > 2m-1$, both μ and the connector term grow, but μ grows faster, widening the gap.

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v_0

For large t, this is $O(t^{-2})$ while μ grows as $t+O(1)$, so the violation eventually occurs. Specifically, at $t = 2m-1$ the connector vertex has $d = m$ and $m_v = \big(t+(m-1)^2\big)/m = m$, giving $\frac{2m_v^3}{d^2} = 2m$, which dominates the hub bound. By Proposition 4.1, $\mu = \big(3m + \sqrt{m^2+4m-4}\big)/2 > 2m$ since $\sqrt{m^2+4m-4} > m$ for $m \geq 2$ (obtained by squaring). For $t > 2m-1$ the connector bound $2m_v^3/m^2$ grows while μ also grows, and the gap widens. □

Theorem 4.3. *BHS Bound 13,*

$$\mu(G) \leq \max_{v \in V} \frac{2m_v^4}{d_v^3}, \tag{6}$$

is violated by $\mathrm{SoC}(K_m, t)$ *for all* $m \geq 5$ *and* $t = 2m-1$. *In particular,* $\mathrm{SoC}(K_9, 19)$ *with* $n = 172$ *satisfies* $\mu(G) \approx 20.685$ *while the bound gives* ≈ 19.845.

Proof. The connector vertex has $d = m$ and $m_v = \big(t + (m-1)^2\big)/m$. At $t = 2m-1$ we have $t + (m-1)^2 = m^2$, giving $B_{13}^{\mathrm{conn}} = 2m^8/m^7 = 2m$. By Proposition 4.1, $\mu = (3m + \sqrt{m^2+4m-4})/2$. Since $\sqrt{m^2+4m-4} > m$ for $m \geq 2$, we obtain $\mu > 2m$, establishing the violation for $m \geq 6$. For $m = 5$

EXPLANATION (ENGLISH)

Think of the bound as a speed limit posted by the formula — Theorem 4.3 is showing a car (the actual spectral radius) that consistently drives faster than the posted limit, proving the sign is set too low. The specific case SoC(K_9, 19) is the "radar-verified" ticket: μ ≈ 20.685 exceeds the bound ≈ 19.845.

Relation to previous sections: Earlier theorems (4.1, 4.2) established μ via the closed form and violated weaker bounds using the hub vertex. Here, Bound 13 uses m_v^4/d_v^3 instead of m_v^3/d_v^2 — a tighter formula — yet the *connector* vertex (not the hub) is the violating witness, showing a different structural role than before.

Even a fourth-power bound on the spectral radius can be defeated by a graph whose connector vertex accumulates edges fast enough that the actual eigenvalue outpaces the formula.

ORIGINAL (ENGLISH)

Proof. The $\mathbb{Z}_2$ symmetry swapping the two hubs decomposes eigenspaces into symmetric and antisymmetric sectors.

Symmetric sector. Quotient matrix $Q_+ = \left(\begin{smallmatrix} k & -k \\ -1 & 1 \end{smallmatrix}\right)$, eigenvalues 0 and $k+1$.

Antisymmetric sector. Quotient matrix $Q_- = \left(\begin{smallmatrix} k+2 & -k \\ -1 & 1 \end{smallmatrix}\right)$, characteristic polynomial $\lambda^2 - (k+3)\lambda + 2 = 0$, eigenvalues $(k+3 \pm \sqrt{k^2+6k+1})/2$.

Pendant leaves contribute eigenvalue 1 with multiplicity $2(k-1)$. Since $\sqrt{k^2+6k+1} > k-1$ for $k \geq 2$, the antisymmetric λ_+ is the global maximum. □

EXPLANATION (ENGLISH)

Proposition 4.4 — Laplacian Spectral Radius of DS(k, k)

This section closes the **SoC bound violation** argument and derives the exact spectral radius for the **double-star DS(k, k)**.

$$\mu\big(DS(k,k)\big) = \frac{(k+3) + \sqrt{k^2+6k+1}}{2}$$

- k: number of pendant leaves attached to each hub
- μ: largest Laplacian eigenvalue (spectral radius)
- Q^+, Q^-: 2×2 quotient matrices from symmetric / antisymmetric decomposition
- λ^+: the dominant eigenvalue from the antisymmetric sector

Proof sketch: The graph's $\mathbb{Z}_2$ symmetry (swapping two hubs) splits eigenvectors into two sectors. The symmetric sector yields eigenvalues 0 and $k+1$; the antisymmetric sector solves $\lambda^2 - (k+3)\lambda + 2 = 0$, giving $\lambda^+ = \frac{(k+3)+\sqrt{k^2+6k+1}}{2}$. Pendant leaves add eigenvalue 1 with multiplicity $2(k-1)$. Since $\sqrt{k^2+6k+1} > k-1$ for $k \geq 2$, λ^+ dominates.

> Think of the two hubs as a seesaw. Symmetric modes have both sides moving together (lower energy); antisymmetric modes have them moving in opposition (higher tension, larger eigenvalue). The largest eigenvalue always lives in the "opposing" antisymmetric mode.

ORIGINAL (ENGLISH)

Proof. The $\mathbb{Z}_2$ symmetry swapping the two hubs decomposes eigenspaces into symmetric and antisymmetric sectors.

Symmetric sector. Quotient matrix $Q_+ = \left(\begin{smallmatrix} k & -k \\ -1 & 1 \end{smallmatrix}\right)$, eigenvalues 0 and $k+1$.

Antisymmetric sector. Quotient matrix $Q_- = \left(\begin{smallmatrix} k+2 & -k \\ -1 & 1 \end{smallmatrix}\right)$, characteristic polynomial $\lambda^2 - (k+3)\lambda + 2 = 0$, eigenvalues $(k+3 \pm \sqrt{k^2+6k+1})/2$.

Pendant leaves contribute eigenvalue 1 with multiplicity $2(k-1)$. Since $\sqrt{k^2+6k+1} > k-1$ for $k \geq 2$, the antisymmetric λ_+ is the global maximum. $\square$

EXPLANATION (ENGLISH)

Connection to previous sections: Earlier sections established interior upper bounds on μ for graph families. The opening lines here confirm those bounds are *violated* by $SoC(K_5, 9)$ ($\mu \approx 10.702 > 10.195$). Proposition 4.4 now provides a **closed-form counterpart** for DS(k,k), enabling direct comparison against those same interior bounds in subsequent analysis.

> The $\mathbb{Z}_2$ symmetry reduction turns a large eigenvalue problem into two 2×2 systems, making an otherwise intractable closed-form result elementary.

$$\mu(G) \le \max_{(i,j)\in E} \left(2 + \sqrt{(d_i - d_j)^2 + 2(d_i m_i + d_j m_j) - 4(m_i + m_j) + 4}\right), \quad (8)$$

is violated by DS(k,k) *for all* $k \ge 6$. *At* $k = 6$, DS$(6,6)$ *with* $n = 14$ *satisfies* $\mu(G) \approx 8.772$ *while the bound gives* ≈ 8.676.

Proof. The hub–leaf edge maximizes Bound 45 for $k \ge 6$. Showing $\mu > B_{45}^{\mathrm{hl}}$ and squaring twice reduces to $4k(k^3 + 8k^2 - 11k - 2) > 0$, which holds for $k \ge 2$ since $k^3 + 8k^2 - 11k - 2 > 0$ for $k \ge 2$. The gap satisfies $B_{45} - \mu = -1/(2k) + O(k^{-2})$. □

Proposition 4.6. *For* $t \ge 2$ *and* $s \ge 1$, *the Laplacian spectral radius of* SoS(t,s) *is*

$$\mu(\mathrm{SoS}(t,s)) = \frac{(t+s+2) + \sqrt{(t-s)^2 + 4s}}{2}. \quad (9)$$

Proof. Equitable partition into {hub}, {intermediates}, {leaves}. The quotient Laplacian is $Q_L = \begin{pmatrix} t & -t & 0 \\ -1 & 1+s & -s \\ 0 & -1 & 1 \end{pmatrix}$ with characteristic polynomial $-\lambda[\lambda^2 - (t+s+2)\lambda + n]$, where $n = 1 + t(1+s)$. Discriminant $(t-s)^2 + 4s > 0$. □

Theorem 4.5 — BHS Bound 45 & Violation by DS(k,k)

BHS Bound 45 gives an upper bound on the Laplacian spectral radius $\mu(G)$ via edge-level degree statistics:

$$\mu(G) \le \max_{(i,j)\in E} \left(2 + \sqrt{(d_i - d_j)^2 + 2(d_i m_i + d_j m_j) - 4(m_i + m_j) + 4}\right) \quad (8)$$

- $\mu(G)$: largest eigenvalue of the Laplacian matrix
- d_i, d_j: degrees of endpoints of edge (i,j)
- m_i, m_j: average degrees of neighbors of i and j

The **hub–leaf edge** achieves the maximum in (8) for $k \ge 6$. Showing $\mu > B_{45}^{hl}$ and squaring twice reduces to $4k(k^3 + 8k^2 - 11k - 2) > 0$, which holds for $k \ge 2$. The asymptotic gap is $B_{45} - \mu = -\frac{1}{2k} + O(k^{-2})$, confirming the bound is tight but violated.

> Think of Bound 45 as a speed-limit sign computed from road widths (degrees) — DS(k,k) is a car that legally exceeds it, proving the sign is miscalibrated for large k.

Relation to prior sections: DS(k,k) was introduced earlier as the candidate counterexample graph family; this theorem confirms it actively breaks Bound 45, motivating the search for tighter bounds.

$$\mu(G) \le \max_{(i,j)\in E} \left(2 + \sqrt{(d_i - d_j)^2 + 2(d_i m_i + d_j m_j) - 4(m_i + m_j) + 4}\right), \quad (8)$$

is violated by $\mathrm{DS}(k,k)$ *for all* $k \ge 6$. *At* $k = 6$, $\mathrm{DS}(6,6)$ *with* $n = 14$ *satisfies* $\mu(G) \approx 8.772$ *while the bound gives* ≈ 8.676.

Proof. The hub–leaf edge maximizes Bound 45 for $k \ge 6$. Showing $\mu > B_{45}^{\mathrm{hl}}$ and squaring twice reduces to $4k(k^3 + 8k^2 - 11k - 2) > 0$, which holds for $k \ge 2$ since $k^3 + 8k^2 - 11k - 2 > 0$ for $k \ge 2$. The gap satisfies $B_{45} - \mu = -1/(2k) + O(k^{-2})$. □

Proposition 4.6. *For* $t \ge 2$ *and* $s \ge 1$, *the Laplacian spectral radius of* $\mathrm{SoS}(t,s)$ *is*

$$\mu\big(\mathrm{SoS}(t,s)\big) = \frac{(t+s+2) + \sqrt{(t-s)^2 + 4s}}{2}. \quad (9)$$

Proof. Equitable partition into {hub}, {intermediates}, {leaves}. The quotient Laplacian is $Q_L = \begin{pmatrix} t & -t & 0 \\ -1 & 1+s & -s \\ 0 & -1 & 1 \end{pmatrix}$ with characteristic polynomial $-\lambda[\lambda^2 - (t+s+2)\lambda + n]$, where $n = 1 + t(1+s)$. Discriminant $(t-s)^2 + 4s > 0$. □

Proposition 4.6 — Exact μ for SoS(t,s)

$$\mu(\mathrm{SoS}(t,s)) = (t+s+2) + \frac{\sqrt{(t-s)^2 + 4s}}{2} \quad (9)$$

Proved via **equitable partition** into {hub}, {intermediates}, {leaves}, yielding the 3×3 quotient Laplacian Q_L whose characteristic polynomial factors out $-\lambda$ leaving a tractable quadratic.

Having a closed-form μ for SoS(t,s) gives a concrete benchmark to test whether any proposed spectral bound is sharp on star-of-stars topology.

Proof. The hub–intermediate edge has $d_h = 39$, $m_h = 9$, $d_w = 9$, $m_w = 47/9$. By Proposition 4.6, $\mu = (49 + \sqrt{993})/2$. The inequality $\mu > B_{40}$ reduces to $\sqrt{993} > 27.25$, i.e., $993 > 742.6$, which holds. □

Remark 4.8 (Bound 48 — degenerate case). BHS Bound 48,

$$\mu(G) \leq \max_{(i,j)\in E} \frac{2(d_i^2 + d_j^2)}{2 + \sqrt{2(d_i^2 + d_j^2) - 4(m_i + m_j) + 4}},$$

is violated by P_3: the bound gives $10/(2+\sqrt{2}) \approx 2.929$ while $\mu(P_3) = 3$. This is the only counterexample found.

5. Why these bounds fail

Section 4.7–4.8 & § 5 Intro: Counterexamples to BHS Bounds 40 and 48

This section delivers the core payoff of the paper: explicit graphs that **violate** two published BHS upper bounds on $\mu(G)$, the spectral radius.

Variable Guide — Bound 40

$$\mu(G) \leq \max_{(i,j)\in E} \frac{1}{2} + \sqrt{\frac{2(m_i - 1)^2 + 2(m_j - 1)^2 + d_i^2 + d_j^2 - d_i m}{2}}$$

- $\mu(G)$: largest eigenvalue (spectral radius) of the adjacency matrix
- d_i, d_j: degrees of endpoints of edge (i, j)
- m_i, m_j: average degrees of neighbors of i and j
- E: edge set of G; bound is the maximum over all edges

Theorem 4.7 proof sketch: For SoS(39, 8) with $n = 352$, the hub–intermediate edge yields $d_h = 39,\ m_h = 9,\ d_w = 9,\ m_w = 47/9$. By Proposition 4.6 (previous section), $\mu = \frac{49+\sqrt{993}}{2} \approx 40.256$, while Bound 40 gives ≈ 38.982. The gap reduces to checking $\sqrt{993} > 27.25$, i.e., $993 > 742.6$. ✓

Remark 4.8: **Bound 48** is violated by the trivial path P_3 (3 vertices): the bound gives $\frac{10}{2+\sqrt{2}} \approx 2.929 < \mu(P_3) = 3$. Notably, this is the *only* counterexample found for Bound 48, suggesting near-tightness.

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Proof. The hub–intermediate edge has $d_h = 39$, $m_h = 9$, $d_w = 9$, $m_w = 47/9$. By Proposition 4.6, $\mu = (49 + \sqrt{993})/2$. The inequality $\mu > B_{40}$ reduces to $\sqrt{993} > 27.25$, i.e., $993 > 742.6$, which holds. □

Remark 4.8 (Bound 48 — degenerate case). BHS Bound 48,

$$\mu(G) \leq \max_{(i,j)\in E} \frac{2(d_i^2 + d_j^2)}{2 + \sqrt{2(d_i^2 + d_j^2) - 4(m_i + m_j) + 4}},$$

is violated by P_3: the bound gives $10/(2+\sqrt{2}) \approx 2.929$ while $\mu(P_3) = 3$. This is the only counterexample found.

5. Why these bounds fail

EXPLANATION (ENGLISH)

Think of these bounds as posted speed limits — Theorem 4.7 is a car clocked faster than the sign allows; Remark 4.8 finds one tiny side-street where the limit is also too low, but everywhere else the limit holds.

Connection to previous sections: Proposition 4.6 computed μ analytically for SoS graphs via a characteristic polynomial; Theorem 4.7 plugs that closed form directly into Bound 40 to turn an inequality check into simple arithmetic. Section 5 ("Why these bounds fail") will now explain the structural reason behind these failures.

Two different BHS bounds fail for opposite reasons: Bound 40 fails on a large, carefully engineered SoS graph, while Bound 48 fails on the simplest possible path — revealing that no single spectral bound is universally tight.

5.1. Hub-invisible bounds

Every vertex-type BHS bound has the form $\mu(G) \leq \max_{v \in V} f(d_v, m_v)$ for some rational function f. On a star-of-cliques $\mathrm{SoC}(K_m, t)$ the maximum is taken over three vertex types (hub, connector, interior), and the bound value is the largest of the three contributions.

Definition 5.1 (Hub-invisible). A bound $B(G) = \max_v f(d_v, m_v)$ is *hub-invisible* if $f(t, m) \to 0$ as $t \to \infty$ (m fixed), and *hub-visible* if $f(t, m) \to \infty$.

Proposition 5.2 (Hub-invisible bounds fail). (a) ***Bounds 11, 13*** *are hub-invisible:* $f_{11} = 2m^3/t^2 \to 0$, $f_{13} = 2m^4/t^3 \to 0$.

(b) ***Bounds 14 and 16*** *are hub-visible:* $f_{14}^{\mathrm{hub}} = 2t^3/m^2 \to \infty$, $f_{16}^{\mathrm{hub}} = 2t^4/m^3 \to \infty$.

5.1 Hub-invisible Bounds

Variable / Symbol Explanation

- $\mu(G)$: the BHS spectral bound being estimated
- d_v: degree of vertex v
- m_v: average neighbor degree (or local graph parameter) at v
- t: hub degree in **SoC**(K_m, t) — grows as $t \to \infty$
- m: clique size (fixed while t varies)

$$\mu(G) \leq \max_{v \in V} f(d_v, m_v)$$

Analogy: Imagine judging a school by its loudest student. A *hub-invisible* bound is like a metric that ignores the loudest student entirely — the louder (higher t) they get, the less the metric registers them, so the bound drifts to zero while the true value stays large.

Concept Relationships — Prior sections established vertex-type BHS bounds on **SoC**(K_m, t) by evaluating $f(d_v, m_v)$ separately for hub, connector, and interior vertices. This section classifies those bounds by whether the *hub vertex's contribution* vanishes or explodes as $t \to \infty$.

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5.1. Hub-invisible bounds

Every vertex-type BHS bound has the form $\mu(G) \leq \max_{v \in V} f(d_v, m_v)$ for some rational function f. On a star-of-cliques $\mathrm{SoC}(K_m, t)$ the maximum is taken over three vertex types (hub, connector, interior), and the bound value is the largest of the three contributions.

Definition 5.1 (Hub-invisible). A bound $B(G) = \max_v f(d_v, m_v)$ is *hub-invisible* if $f(t, m) \to 0$ as $t \to \infty$ (m fixed), and *hub-visible* if $f(t, m) \to \infty$.

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Proposition 5.2 summary:

— Hub-invisible (fail): $f_{11} = \frac{2m^3}{t^2} \to 0$, $f_{13} = \frac{2m^4}{t^3} \to 0$ — hub suppressed.

— Hub-visible (valid): $f_{14}^{\mathrm{hub}} = \frac{2t^3}{m^2} \to \infty$, $f_{16}^{\mathrm{hub}} = \frac{2t^4}{m^3} \to \infty$, $f_{30}^{\mathrm{hub}} \sim \frac{t^2}{m} \to \infty$.

> A bound that cannot "see" the hub vertex will always underestimate μ on hub-dominated graphs, making hub-invisibility a decisive structural failure criterion.

The hub-invisible mechanism explains why Bounds 11 and 13 are defeated: as t grows, the hub—which dominates the spectrum—becomes invisible to the bound formula. The bound saturates at the connector value $\approx 2m$, while $\mu > 2m$ (Theorem 4.2).

Remark 5.3. Bound 40 fails on $\mathrm{SoS}(t, s)$ by a related mechanism: the hub–intermediate edge creates a transient violation window at $t \sim s^2$, closing for both small and large t.

5.2. Edge-based antisymmetric modes

Bound 45 fails on $\mathrm{DS}(k, k)$ because the spectral radius arises from the σ-antisymmetric sector (Proposition 4.4), where the two hubs carry opposite signs. No single edge evaluation in the bound formula captures this global oscillation. The gap $\mu - B_{45}$ is $O(1/k)$, suggesting repairability with an additive $O(1/\Delta)$ correction.

5.1–5.2 Why Certain Bounds Fail: Hub-Invisibility and Antisymmetric Modes

Connection to previous sections: Sections 4.x established the spectral radius μ of hub-based graphs (SoS, DS) and derived Theorem 4.2 ($\mu > 2m$). This section now explains *mechanistically* why existing bounds (B11, B13, B40, B45) cannot detect that gap — tying the counterexamples back to structural blind spots in the bound formulas.

Variable / Symbol Reference

- μ: spectral radius of the graph (leading eigenvalue)
- m: connector/pendant edge count; $\approx 2m$ is the saturation ceiling of the defeated bounds
- t, s: growth parameters of $\mathrm{SoS}(t, s)$; $t \sim s^2$ marks the transient violation window
- k: hub degree in $\mathrm{DS}(k, k)$; gap $\mu - B_{45} = O(1/k)$
- Δ: maximum degree; $O(1/\Delta)$ is the proposed additive correction scale

> **Analogy — Hub-Invisibility:** Imagine a loudhailer (the hub) dominating a room's noise level, but the sound meter only measures wall outlets (connectors). As the room grows, the meter still reads the outlet level $\approx 2m$, completely missing the loudhailer. That's exactly how B11/B13 saturate while μ keeps growing.

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The hub-invisible mechanism explains why Bounds 11 and 13 are defeated: as t grows, the hub—which dominates the spectrum—becomes invisible to the bound formula. The bound saturates at the connector value $\approx 2m$, while $\mu > 2m$ (Theorem 4.2).

Remark 5.3. Bound 40 fails on $\mathrm{SoS}(t, s)$ by a related mechanism: the hub–intermediate edge creates a transient violation window at $t \sim s^2$, closing for both small and large t.

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EXPLANATION (ENGLISH)

$$\mu > 2m, \quad \text{but } B_{11}, B_{13} \to 2m \text{ as } t \to \infty$$

For **Bound 40** on $\mathrm{SoS}(t, s)$, the failure is *transient*: the hub's intermediate edge creates a violation only near $t \sim s^2$, closing at both extremes — a narrow structural window the bound misses.

For **Bound 45** on $\mathrm{DS}(k, k)$, the failure is *modal*: μ comes from the **σ-antisymmetric sector** (Proposition 4.4), where the two hubs carry opposite signs. A single-edge evaluation in B45 sees only local magnitudes, not the global oscillation pattern. Crucially, the gap is $O(1/k)$ — small and repairable by an additive $O(1/\Delta)$ correction.

> The two failure modes reveal a design principle: bounds built on local edge evaluations are structurally blind to either hub dominance (scale invisibility) or hub sign-cancellation (modal invisibility), and the repair strategy differs accordingly.

5.3. Characterizing vulnerable bounds

Conjecture 5.4. *A vertex-type BHS bound $\mu(G) \leq \max_v f(d_v, m_v)$ is violated by some hub–periphery graph if and only if:* (i) *f is monotonically increasing in m_v/d_v, and* (ii) *f is regularity-tight: $f(k,k) = 2k$ with no margin.*

6. Classification of the remaining 33 bounds

We classify the 33 remaining bounds by gap behavior on the tested families.

The SUPER-SAFE bounds include all hub-visible bounds (14, 16, 30). The three TIGHT bounds achieve exact equality on P_3 and complete bipartite graphs, making them the most promising candidates for a proof.

7. Discussion

Summary.. BHS [2] proposed 68 conjectured bounds for $\mu(G)$. Ghebleh et al. [4] disproved 30 via cross-entropy methods. We have disproved 5 more by structural construction, leaving 33 open.

5.3 · 6 · 7 — Vulnerable Bounds, Classification & Discussion

Conjecture 5.4 identifies exactly when a vertex-type BHS bound can be broken by a hub-periphery graph.

$$\mu(G) \leq \max_v f(d_v,\, m_v)$$

- $\mu(G)$: spectral radius of graph G
- d_v: degree of vertex v
- m_v: average neighbor degree of v
- $f(d_v, m_v)$: the bounding function evaluated per vertex

Two conditions must *both* hold for vulnerability:

- **(i)** f is monotonically increasing in the ratio m_v/d_v — rewards high-degree hubs
- **(ii)** f is **regularity-tight**: $f(k,k) = 2k$ with zero margin on regular graphs

> Think of condition (ii) like a bridge rated for exactly its maximum load — no safety margin means any asymmetric stress (a hub-periphery structure) causes failure immediately.

Section 6 classifies the 33 surviving bounds by gap behavior:

- Super-safe bounds : hub-visible bounds (indices 14, 16, 30) — never violated
- Tight bounds : achieve exact equality on P_3 and complete bipartite graphs — top proof candidates

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5.3. Characterizing vulnerable bounds

Conjecture 5.4. *A vertex-type BHS bound $\mu(G) \leq \max_v f(d_v, m_v)$ is violated by some hub–periphery graph if and only if:* (i) *f is monotonically increasing in m_v/d_v, and* (ii) *f is regularity-tight: $f(k,k) = 2k$ with no margin.*

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Summary.. BHS [2] proposed 68 conjectured bounds for $\mu(G)$. Ghebleh et al. [4] disproved 30 via cross-entropy methods. We have disproved 5 more by structural construction, leaving 33 open.

EXPLANATION (ENGLISH)

Connecting to prior sections: Section 5's hub-periphery constructions provided the counterexamples that filter these categories. Bounds that survived those constructions land in the super-safe or tight classes here.

Section 7 — Running tally of the 68 BHS conjectures:

By combining cross-entropy disproof (prior work) with structural hub-periphery constructions (this paper), 35 of 68 bounds are now eliminated — leaving a well-characterized open set of 33.

ORIGINAL (ENGLISH)

Structural vs. computational approaches.. Cross-entropy search explores adjacency matrices for $n \leq 24$. Our smallest counterexample has $n = 29$, beyond exhaustive reach. Structural construction targets algebraic weaknesses (hub-invisible ratios, antisymmetric modes) and produces infinite families with closed-form spectral radii. The two approaches are complementary.

Gap asymptotics.. Bounds 11/13: gap diverges (irreparable). Bound 40: transient window (max gap 1.74). Bound 45: gap $\to 0$ as $1/(2k)$ (repairable). Bound 48: degenerate P_3 case only.

EXPLANATION (ENGLISH)

Structural vs. Computational Approaches & Gap Asymptotics

Cross-entropy search is the computational arm: it blindly samples adjacency matrices up to $n \leq 24$. The **structural construction** arm is algebraic — it exploits known weaknesses (hub-invisible ratios, antisymmetric modes) to manufacture infinite families analytically.

> Think of it like cracking a safe: the computational approach tries random combinations until something clicks (works for small safes), while the structural approach studies the lock's mechanism to craft a master key valid for any size.

Connection to earlier sections: The hub-invisible ratios and antisymmetric modes were identified as the algebraic vulnerabilities when each bound was originally derived; here they become the *entry points* for targeted construction, bridging the earlier theoretical analysis to concrete counterexamples.

Gap Asymptotics Summary

- **Bounds 11/13:** gap diverges — counterexamples grow unboundedly worse (*irreparable*)
- **Bound 40:** gap peaks at 1.74 then closes — a *transient* failure window
- **Bound 45:** gap $\to 0$ as $\frac{1}{2k}$ — asymptotically tight (*repairable*)
- **Bound 48:** only the degenerate P_3 graph fails

> The gap asymptotics classify each bound's severity: diverging gaps expose fundamentally flawed bounds, while vanishing gaps suggest the bound needs only a small correction term.

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Open questions.. (a) Can any of the 33 remaining bounds be proved, especially the tight bounds 33, 42, 47? (b) Do three-level hierarchies violate additional bounds? (c) Can Conjecture 5.4 be proved? (d) Can Bound 45 be repaired with an $O(1/\Delta)$ correction?

Declaration of Generative AI in Scientific Writing

During the preparation of this work, the author used Claude (Anthropic; model: Claude Opus 4, 2025) for computational assistance, including code development for graph construction and eigenvalue computation, structural analysis of graph families, and systematic counterexample search. The author reviewed and edited all content and takes full responsibility for the publication.

EXPLANATION (ENGLISH)

Open Questions & AI Declaration

This closing section surfaces **unresolved conjectures** and discloses the computational tools used — standard practice in modern graph theory papers.

- **Bound 33, 42, 47**: tight bounds still awaiting proof
- **Three-level hierarchies**: unexplored extension of the two-level containment structure analyzed throughout
- **Conjecture 5.4**: a specific structural claim left open
- **Bound 45 + O(1/Δ) correction**: Δ = maximum degree; the bound may need a degree-dependent repair term

> Open questions are like a map's blank edges — the known territory (proved bounds) is charted, but these four corners remain unexplored, inviting future expeditions.

Relationship to previous sections: The 33 remaining bounds directly reference the full table of spectral/structural inequalities built up across Sections 2–5. Conjecture 5.4 and Bound 45 are specific entries from that hierarchy; the three-level question extends the two-level containment lattice established earlier.

The **AI declaration** is noteworthy: Claude Opus 4 assisted with graph construction, eigenvalue computation, and counterexample search — all computational tasks. Human authorship retained full responsibility for mathematical reasoning and conclusions.

> The open questions map exactly onto the paper's two hard problems: proving tight bounds and understanding deeper hierarchy violations — the work is complete but intentionally leaves the frontier visible.

ORIGINAL (ENGLISH)

[1] A. E. Brouwer, W. H. Haemers, Spectra of Graphs, Universitext, Springer, 2012. doi:10.1007/978-1-4614-1939-6.

[2] V. Brankov, P. Hansen, D. Stevanović, Automated conjectures on upper bounds for the largest Laplacian eigenvalue of graphs, Linear Algebra and its Applications 414 (2–3) (2006) 407–424. doi:10.1016/j.laa.2005.10.017.

[3] G. Caporossi, P. Hansen, Variable neighborhood search for extremal graphs. 1. The AutoGraphiX system, Discrete Mathematics 212 (1–2) (2000) 29–44. doi:10.1016/S0012-365X(99)00206-X.

EXPLANATION (ENGLISH)

Acknowledgements & References

This closing section credits human collaborators, computational tools, and foundational literature that underpin the paper's results.

Concept Relationships: These references directly anchor the methodology used throughout the paper — **BHS bound list** (cited from Brankov–Hansen–Stevanović, [2]) was the benchmark against which new bounds were tested; **AutoGraphiX** ([3]) represents the automated conjecture framework that contextualizes this work's approach.

- **[1] Brouwer & Haemers (2012)** — standard reference text for spectral graph theory, supplying foundational definitions used in earlier sections
- **[2] Brankov, Hansen, Stevanović (2006)** — source of the BHS upper-bound conjectures on the largest Laplacian eigenvalue μ_1
- **[3] Caporossi & Hansen (2000)** — AutoGraphiX system for automated extremal graph conjectures, predecessor methodology

The acknowledgement that "all mathematical results were independently verified by the author" is like a pilot confirming the autopilot's route by hand — AI assistance (Claude/Anthropic) accelerated computation, but human proof remains the authoritative check.

Citing the BHS list alongside independent verification signals that the paper's contributions are positioned as improvements over an established, publicly reproducible baseline.

ORIGINAL (ENGLISH)

[4] M. Ghebleh, S. Al-Yakoob, A. Kanso, D. Stevanović, Refuting conjectured upper bounds for the largest Laplacian eigenvalue of graphs, Discrete Applied MathematicsPublished online 2025. Available at https://arxiv.org/abs/2403.18429 (2025).

[5] A. Z. Wagner, Constructions in combinatorics via neural networks, arXiv preprint arXiv:2104.14516 (2021).

[6] N. Bouffard, J. Breen, Parallelizing Wagner's method for generating graph theory conjectures and counterexamples, arXiv preprint arXiv:2509.01607 (2025).

[7] M. Roucairol, T. Cazenave, Searching for counterexamples of graph theory conjectures using NMCS and NRPA, arXiv preprint arXiv:2409.18626 (2024).

EXPLANATION (ENGLISH)

References [4]–[7]: AI & Computational Methods for Graph Theory Conjectures

This block continues the reference list, grouping works on **automated conjecture refutation** and **neural/search-based counterexample generation** — the computational backbone cited throughout the paper's methodology.

- **[4] Ghebleh et al. (2025)**: Directly refutes conjectured upper bounds for the largest Laplacian eigenvalue — closely aligned with this paper's own conjecture-testing goals.
- **[5] Wagner (2021)**: Pioneered using neural networks to construct combinatorial counterexamples; foundational for [6] and [7].
- **[6] Bouard & Breen (2025)**: Parallelizes Wagner's method, scaling up graph conjecture search.
- **[7] Roucairol & Cazenave (2024)**: Applies NMCS/NRPA (nested Monte Carlo search) as an alternative search strategy.

> Wagner's method is like teaching a dog to fetch counterexamples: train it with rewards when it finds violations. [6] puts many dogs to work simultaneously; [7] uses a smarter search map instead.

Relationship to previous sections: These references underpin the paper's experimental comparisons — where the authors benchmark their approach against or build upon Wagner-style and Monte Carlo search methods.

> The convergence of Laplacian spectral theory [4] with neural and Monte Carlo search [5–7] shows the field is rapidly automating mathematical discovery itself.

ORIGINAL (ENGLISH)

[8] C. Godsil, G. Royle, Algebraic Graph Theory, Vol. 207 of Graduate Texts in Mathematics, Springer, 2001. doi:10.1007/978-1-4613-0163-9.

EXPLANATION (ENGLISH)

This is a **bibliography entry** for a foundational algebraic graph theory textbook used to support the theoretical framework throughout this paper.

Like citing a dictionary — the paper borrows standard definitions and theorems on graph spectra and adjacency matrices from this reference.

This reference anchors the graph-theoretic foundations underpinning the spectral or structural analyses presented in earlier sections.